

2080

Bachelor of Science in Computer Science and Information Technology

Course Title: **Data Warehousing and Data Mining**

Code No: **CSC410** Semester: **7th Semester**

Full Marks: **60** Pass Marks: **24** Time: **dia data mining along with the concept of similarity search.12.Write down short notes ona. Support Vector Machineb. Multi-dimensional Data Model**

Candidates are required to answer the questions in their own words as far as practicable.

Section A

1. State Apriori property. Find frequent itemsets and association rules from the transaction database given below using Apriori algorithm. Assume min. support is 50% and min confidence is 75%.
Transaction IDItems
Purchased1Juice2Juice3Bread, Butter4Juice, Milk5Cheese, Juice, Milk
2. How classification differs from regression. Train ID3 classifier using the dataset given below. Then predict class label for the dataAgeCompetitionTypeProfit (Class Label)oldYesSWDownoldNoSWDownoldNoHWDownMidYesSWDownMidYesHWDownMidNoHWUpMidNoSWUpNewYesSWUpNewNoHWUpNewNoSWUp
3. Why the concept of data mart is important? Discuss different data warehouse scheme with examples.

Section B

Attempt any Eight questions[8×5=40]

4. How OLAP differ from data mining? Explain various stages of KDD with suitable block diagram.
5. Discuss different ways of smoothing noisy data along with suitable examples.
6. How many cuboids are possible from 5-dimensional data? Discuss the concept of full cube and iceberg cube.
7. How K-medoids clustering differs from K-means clustering? Divide the following data points into two clusters using k-medoids algorithm. Show computation upto 2 iterations.{(70,85), (65,80), (72,88), (75,90), (60,50), (64,55), (62,52), (63,58)}
8. Discuss working of DBSCAN algorithm.
9. Which algorithm is used for training multi-layer perceptron? Discuss the algorithm in detail.
10. Explain the OLAP operations with example.
11. Discuss the concept of multimedia data mining along with the concept of similarity search.
12. Write down short notes ona. Support Vector Machineb. Multi-dimensional Data Model